

EE 516: Biomedical Systems

Course Code	EE 516
Course Name	Biomedical Systems
L-T-P-C	2.5-1.5-0-4
Prerequisites	IC 110 Engineering Mathematics (IC110), IC 161 Applied Electronics
Intended for	UG/PG
Approval	9th Senate (May 2016)

Course Contents

- **Introduction to Biomedical Systems:** Introduction to System Science - Notion of dynamic systems: modeling and simulation using MATLAB - Biomedical systems as dynamic systems - Compartmental modeling of biological systems - Eye movement model - Muscle model - Classical system identification. [4 Lectures]
- **Anatomy and Physiology:** Introduction-Cellular organization Tissues - Major organs and systems - Homeostasis. [6 Lectures]
- **Modelling of human body:** Cardiovascular model, Lung model, Nervous System model, Muscular system model. [5 Lectures]
- **Biomedical sensing:** Bioelectric phenomena - Origin of bio-potentials - Bio-potential measurements ECG, EEG, EMG, ERG, ENG Notion of system identification - Chemical biosensors electrochemical sensors and chemical fibro-sensors - Notion of ion selective field effect transistor (ISFET) and immunologically sensitive field effect transistor (IMFET) - Fundamentals of light propagation in biological tissue - Biophysical measurement techniques using light photoplethysmography - Acoustic biosensors phonocardiography - Photo-acoustic bio-signals estimation of blood glucose. [11 Lectures]
- **Bio-signal processing:** Characterization of bio-signals morphological, statistical and transform features - Frequency domain representation of bio-signals - Noise characteristics - Noise reduction by Ensemble Averaging and Linear Time Invariant A Posteriori - filtering techniques - Signal averaging - Wavelet transform - Compression of bio-signals - lossless and lossy compression. [6 Lectures]
- **Biomedical embedded systems:** Choice of embedded core - Notion of Internet of Things as extended to biomedicine - Embedded processing for disease diagnosis Wearable biomedical embedded systems - Point of care testing devices - Diagnostic processing for detection and classification of diseases - Computational intelligence techniques for disease diagnosis - Classification of cardiac, neuromuscular, neurological and hematological diseases - Memory management issues for diagnostic processing - Power reduction techniques in diagnostic systems. [8 Lectures]
- **Moral and ethical issues in developing Biomedical Systems:** Morality and ethics - Two moral norms: beneficence and nonmaleficence - Human experimentation - Regulation of medical device innovation - Ethical issues in feasibility studies - Ethical issues in treatment use. [2 Lectures]

Course Project

A 6 weeks project where the student will develop a practical biomedical system going through the problem formulation, analysis, design and development phases.

Text Books

1. J. Enderle, S. Blanchard, J. Bronzino, **Introduction to Biomedical Engineering**, Elsevier Academic Press, 2009.
2. R. Begg, D.T.H. Lai, M. Palaniswami, **Computational Intelligence in Biomedical Engineering**, CRC Press, 2008.

References

1. L. Sornmo, P. Laguna, **Bioelectrical Signal Processing in Cardiac and Neurological Applications**, Elsevier Academic Press, 2005.
2. J.G. Webster, **Medical Instrumentation: Application and Design**, John Wiley and Sons, 2003.